MSc. in Computing
Practicum Approval Form

Section 1: Student Details

Project Title:	Walkers LLP Practicum
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Chosen major:	Natural Language Processing
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Date of Submission	24 November, 2024

Section 2: About your Practicum

Please answer all questions below. Please pay special attention to the word counts in all cases.

What is the topic of your proposed practicum? (100 words)

The proposed practicum focuses on text-to-symbolic conversion for legal database queries, leveraging large language models (LLMs) and symbolic reasoning techniques to address challenges in the legal domain. The goal is to develop a system that translates natural language legal questions into structured queries (e.g., SQL or Prolog) for retrieving and reasoning over legal texts, statutes, or case law. By fine-tuning LLMs on domain-specific datasets and integrating logic-based languages, the practicum aims to improve access to legal information, enhance decision-making, and address limitations such as ambiguity and scalability in current legal AI systems.

Please provide details of the papers you have read on this topic (details of 5 papers expected).

1. “How Does NLP Benefit Legal System: A Summary of Legal Artificial Intelligence”: This paper discusses the transformative potential of NLP in the legal domain, including automating document review and legal research, predicting case outcomes, and summarising law documents. It highlights datasets like Supreme Court databases and legislative records, which are used to train ML models for legal applications.

2. “Case-Level Prediction of Motion Outcomes in Civil Litigation”: This paper introduces ML methods for predicting motion outcomes in civil litigation. It uses Connecticut Judicial Branch data and complaint documents converted via OCR. The study employs NLP methods such

as word embeddings (Word2Vec, TF-IDF) and rule-based algorithms. Classifiers include decision trees, Adaboost, and SVM.

3. ”Next-Generation Database Interfaces: A Survey of LLM-based Text-to-SQL”: The paper surveys how large language models (LLMs) enhance the text-to-SQL task, aiming to translate natural language queries into SQL for complex databases. The survey contextualises LLM advancements by comparing them with earlier text-to-SQL approaches, including rule-based systems, deep learning, and pre-trained language models, demonstrating the evolution and gaps in the field.

4. “ From Natural Language to SQL: Review of LLM-based Text-to-SQL Systems”: The paper explores advancements in text-to-SQL systems, focusing on converting natural language queries into accurate SQL for complex databases. It reviews various methodologies, including rule-based systems, deep learning, pre-trained models, and large language models (LLMs), with an emphasis on schema linking and query generation. The findings highlight that LLMs significantly enhance performance through contextual understanding but still face challenges such as ambiguity, scalability, and efficiency. Related work examines the progression of text-to-SQL methods and the role of benchmarks and datasets in addressing research gaps.

5. “Connecting Symbolic Statutory Reasoning with Legal Information Extraction”: The paper addresses the challenge of connecting statutory reasoning with automated legal information extraction by translating natural language case descriptions into structured Prolog knowledge bases (KBs). It proposes an improved ontology for annotating legal cases and adapts state-of-the-art information extraction (IE) models to extract structured representations from the SARA dataset. The study finds that enhanced IE performance correlates with improved accuracy in statutory reasoning tasks, demonstrating the utility of structured representations.

How does your proposal relate to existing work on this topic described in these papers? (200 words)

Our proposal relates closely to the existing work outlined in these papers by addressing similar challenges and leveraging advancements in Legal NLP and text-to-symbolic conversion. Papers such as “How Does NLP Benefit Legal System” and “Case-Level Prediction of Motion Outcomes in Civil Litigation” provide insights into how NLP automates legal tasks, including classification and prediction, which align with our goal of translating legal questions into structured queries. The surveys “Next-Generation Database Interfaces” and “From Natural Language to SQL” directly inform our text-to-SQL methodology, highlighting LLMs’ contextual capabilities while addressing issues like scalability and ambiguity. Lastly, “Connecting Symbolic Statutory Reasoning with Legal Information Extraction” emphasises the integration of logic-based systems like Prolog, which informs our exploration of using these languages for legal reasoning.

In our project, we build upon these works by fine-tuning LLMs on legal datasets to convert natural language into SQL or Prolog queries, creating a pipeline for querying legal databases. Challenges such as dataset availability, discussed in the surveyed works, guide our decision to explore synthetic data generation and fine-tuning strategies. Our evaluation framework also reflects their methodologies, focusing on query accuracy and reasoning performance. This integration synthesises key contributions from prior research into a cohesive system for computational legal tasks.

What are the research questions that you will attempt to answer? (200 words)

Our research aims to address several critical questions related to integrating natural language processing (NLP) and computational law for automating legal queries.

1. How can natural language legal queries be accurately translated into structured symbolic representations like SQL or Prolog for database interaction? This involves exploring how LLMs can be fine-tuned to effectively handle the complexities of legal language and query formulation.
2. What role do datasets and pre-trained models play in improving the accuracy and efficiency of text-to-symbolic conversion in the legal domain? We aim to assess the impact of domain-specific datasets (e.g., MultiLegalPile, LexGLUE) and synthetic data.
3. How can logic-based languages like Prolog be effectively used to encode legal statutes for automated reasoning? This includes evaluating Prolog's ability to handle legal reasoning tasks and its integration with NLP-driven query systems.
4. What are the limitations of LLMs in legal applications, and how can these be mitigated? We will explore strategies such as adapting existing datasets, synthetic data augmentation, and robust evaluation metrics to overcome these challenges.
5. How can AI-driven text-to-symbolic systems streamline legal workflows and reduce manual effort in tasks like legal research, case analysis, and statute compliance?

How will you explore these questions? (Please address the following points. Note that three or four sentences on each will suffice.)

- What software and programming environment will you use?

We will use Python as the primary programming environment, leveraging libraries such as Hugging Face's transformers for model fine-tuning and frameworks like TensorFlow or PyTorch for deep learning. Tools like SQLAlchemy and Prolog engines (e.g., SWI-Prolog) will handle database and symbolic reasoning tasks.

- What coding/development will you do?

We will fine-tune large language models (e.g., GPT, T5) to convert natural language queries into SQL or Prolog. Rule-based logic will be implemented to parse queries for symbolic reasoning. Development will also include building a chatbot interface that integrates query generation, database interaction, and result retrieval. Frontend will be using JavaScript, React.js, Tailwind Css and we will store our data in MongoDB as an industry standard application.

- What data will be used for your investigations?

We will use datasets such as MultiLegalPile, Legal-ML-Datasets, and Spider, which include legal texts, statutes, and text-to-SQL query pairs. Synthetic data may be generated if specific litigation datasets are unavailable.

- Is this data currently available, if not, where will it come from?

Most of the required datasets are publicly accessible through platforms like Hugging Face or research archives. The Spider dataset, for example, is available and can be adapted for legal queries, although its size may pose limitations. For datasets that are unavailable, we plan to create them manually by extracting relevant data from various legal case studies.

- What experiments do you expect to run?

We will test the models on converting natural language queries into SQL or Prolog, evaluating their ability to retrieve relevant results from a legal database. Different datasets will be used to test generalisability.

- What output do you expect to gather?

The output will include generated SQL or Prolog queries and corresponding query results from the database. Logs will capture errors and misclassifications. The ultimate goal is to provide users with precise and reliable results.

- How will the results be evaluated?

We will evaluate the models based on accuracy, precision, recall, and logical consistency of the generated queries. Database interaction performance and retrieval relevance will also be measured. Adjustments will be made iteratively based on these metrics. Feedback from professionals during system testing and demo runs will provide us with valuable insights and to make needed improvements.